

Lecture 02: Electron as a Wave

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Last lecture: electrons as particles

- ▶ Late 1800s: electrons discovered as discrete charged particles with definite mass
- ▶ Classical expectation: electrons travel in straight lines, scatter like billiard balls
- ▶ Particle model predicts: conductivity (Drude model), transparency (plasma frequency), and resonance (cyclotron frequency)

If electrons are particles, their behavior should be entirely predictable from classical mechanics

Maxwell's equations (general)

With the definitions $\mathbf{D} = \varepsilon \mathbf{E}$ ($\mathbf{D}$ = displacement field) and $\mathbf{B} = \mu \mathbf{H}$ ($\mathbf{H}$ = magnetic field),

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

EM waves in conductors: the wave equation

Notation: $\mathbf{H}$ = magnetic field intensity, $\mathbf{J}$ = current density, $\mathbf{D}$ = displacement field, σ = conductivity, ε = permittivity, μ = permeability

- ▶ Maxwell's equations with conduction current:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t}$$

- ▶ For good conductors at moderate frequencies: $\sigma \gg \omega \varepsilon$
(conduction dominates over displacement current)
- ▶ Wave equation becomes diffusion-like, not wave-like

Diffusion-like behavior of electromagnetic waves in conductors:

$$\nabla^2 \mathbf{E} = \mu \sigma \frac{\partial \mathbf{E}}{\partial t} \tag{1}$$

Skin depth in conductors

Notation: k = complex wave vector, δ = skin depth, ω = angular frequency

- ▶ Assume plane wave $\mathbf{E} = \mathbf{E}_0 \exp(ikz - i\omega t)$ propagating into conductor ($z > 0$)
- ▶ Substituting into wave equation:

$$k^2 = i\omega\mu\sigma \quad (2)$$

$$k = \frac{1+i}{\delta} \quad \text{where} \quad \boxed{\delta = \sqrt{\frac{2}{\omega\mu\sigma}}} \quad (3)$$

- ▶ Complex k gives both propagation and exponential decay
- ▶ Real part: oscillation; Imaginary part: attenuation

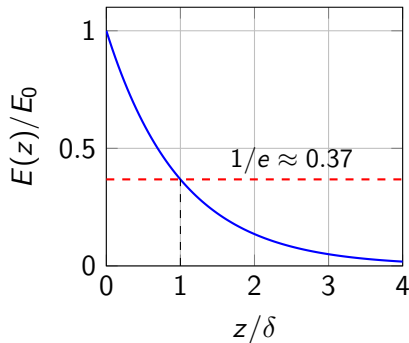
Exponential decay of EM fields in conductors

- ▶ Field amplitude:

$$E(z) = E_0 e^{-z/\delta}$$

decays to $1/e$ at depth δ

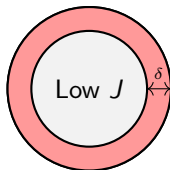
- ▶ Power density decays as $e^{-2z/\delta}$:
energy confined to thin surface layer
- ▶ Higher frequency $\rightarrow$ smaller $\delta \rightarrow$ fields
confined closer to surface



Electromagnetic shielding in conductors

- ▶ Current flows only in skin layer of thickness δ : effective cross-section reduced

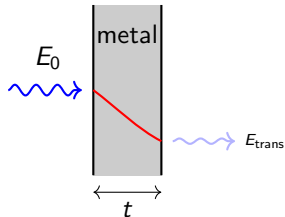
⇒ Use hollow conductors at high frequency!



Current in skin layer

- ▶ Shielding effectiveness is proportional to the thickness t :

$$SE \propto (t/\delta) \text{ dB}$$



Plasma frequency

Notation: ω_p = plasma frequency, N_e = electron density, e = elementary charge, m = electron mass, ε = dielectric permittivity

- ▶ Plasma frequency ω_p marks transition between **opaque** and **transparent** regimes
- ▶ Arises from collective oscillation of free electrons against fixed ion background
- ▶ Depends only on electron density — intrinsic material property

Plasma frequency:

$$\boxed{\omega_p = \sqrt{\frac{N_e e^2}{m \varepsilon}}} = \sqrt{\frac{\sigma}{\varepsilon \tau}} \quad (4)$$

$\omega < \omega_p$	$\omega > \omega_p$
Metal opaque Waves reflected/absorbed	Metal transparent Waves propagate

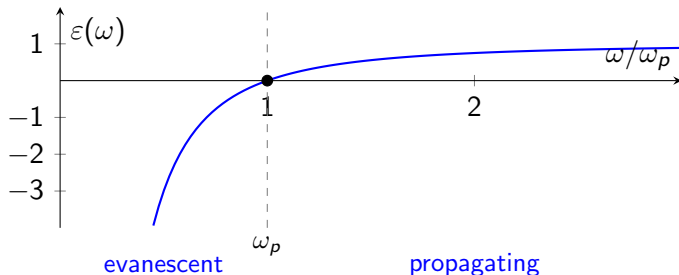
Dielectric constant and wave propagation

Notation: $\varepsilon(\omega)$ = effective dielectric constant, ω_p = plasma frequency, k = wave vector

Effective dielectric constant

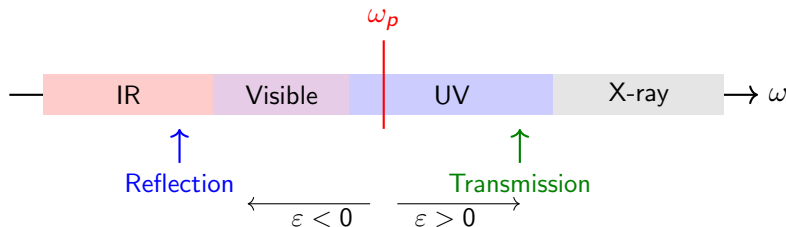
$$\varepsilon_{\text{eff}}(\omega) = 1 - \frac{\omega_p^2}{\omega^2} \quad (5)$$

- ▶ For $\omega < \omega_p$: $\varepsilon < 0$, k is imaginary $\rightarrow$ **evanescent wave**
- ▶ For $\omega > \omega_p$: $\varepsilon > 0$, k is real $\rightarrow$ **propagating wave**



Metallic reflectivity across the EM spectrum

- ▶ Metals **reflect** EM waves when $\omega < \omega_p$ (visible light for most metals)
- ▶ Metals become **transparent** when $\omega > \omega_p$ (UV regime)
- ▶ Typical metal: $\omega_p \approx 2.2 \times 10^{15}$ Hz corresponds to UV wavelength ~ 140 nm



Particles or waves? electromagnetic fields behave as waves, but the photoelectric effect showed that photons also behave as particles.

So far, electrons have been treated as particles. What cast a shadow on this major assumption?

A timeline of the electron problem (so far)

- ▶ 1900 Planck: energy quantization explains blackbody radiation
- ▶ 1905 Einstein: proposes quantization of photons (photoelectric effect)
- ▶ 1911 Rutherford: discovery of nucleus
- ▶ 1913 Bohr: quantized angular momentum levels of energy of the electron in the atom
- ▶ 1923 Compton: scattering and “electron wavelength”
- ▶ 1924 de Broglie: matter waves

Electron as a wave

- ▶ Classical particle model fundamentally incomplete for electrons
- ▶ Wave properties essential to explain diffraction and interference phenomena
- ▶ Key question: **What wavelength should we assign to an electron?**

Particle Picture	Wave Picture
mass m	wavelength $\lambda = ?$
velocity v	frequency $\nu = ?$
momentum $p = mv$	wavenumber $k = ?$

de Broglie's Hypothesis

- ▶ Louis de Broglie (1924): If light waves have particle properties, particles must have wave properties
- ▶ Every particle with momentum p has an associated wavelength
- ▶ This wavelength determines diffraction and interference behavior

analogy with light

Light: wave $\rightarrow$ particle (photon)

Electron: particle $\rightarrow$ wave (matter wave)

de Broglie wavelength

- ▶ For photons: $E = h\nu$ and $E = pc$ combine to give $\lambda = h/p$
- ▶ de Broglie extended this to massive particles: momentum $p = mv$
- ▶ Result: wavelength inversely proportional to momentum

de Broglie wavelength

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

Notation: $h = 6.6 \times 10^{-34}$ J·s (Planck's constant), m = particle mass, v = velocity

Interpretation: momentum-wavelength

- ▶ Higher momentum $\rightarrow$ shorter wavelength (faster particles are more “particle-like”)
- ▶ Lower momentum $\rightarrow$ longer wavelength (slower particles show stronger wave effects)
- ▶ Wave nature becomes observable when λ approaches physical length scales

Fast electron

large p
small λ



particle-like

Slow electron

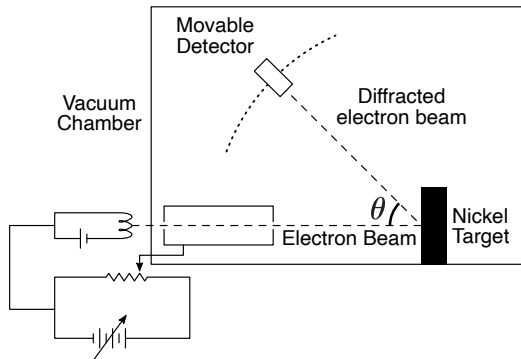
small p
large λ



wave-like

Davisson-Germer experiment

- ▶ 1927: Davisson and Germer scattered electrons off crystalline nickel surface
- ▶ Observed unexpected intensity peaks at specific angles—not predicted by particle model
- ▶ Accidental annealing created large crystal domains enabling coherent scattering
- ▶ Experimental demonstration that electrons exhibit wave behavior



Bragg diffraction

Mechanism: Waves scattered from adjacent crystal planes

- ▶ Path difference between planes separated by d at angle θ : $\Delta = d \sin \theta$
- ▶ Constructive interference requires path difference = integer wavelengths

Condition for Bragg diffraction

$$n\lambda = d \sin \theta \quad (6)$$

Notation: n = diffraction order (integer), d = lattice spacing, θ = scattering angle

Reminder: Diffraction maxima occur only at specific angles determined by λ and d

Measuring the de Broglie wavelength

Experimental verification strategy:

1. Measure diffraction angle θ with known spacing d between crystallographic planes
2. Calculate λ from Bragg condition: $\lambda = \frac{d \sin \theta}{n}$
3. Independently determine electron velocity v from accelerating voltage
4. Compare with de Broglie prediction:

de Broglie Wavelength

$$\lambda = \frac{h}{mv} \quad (7)$$

Checkpoint: Davisson-Germer and Bragg Diffraction

Conceptual questions:

1. **Q1:** The discovery of Davisson-Germer happened because the sample was accidentally heated up and annealed for some time. Why do you think this is important for the discovery?
2. **Q2:** If lattice spacing doubles ($d \rightarrow 2d$), how must electron energy change to observe the same diffraction angle?
3. **Q3:** What would Davisson-Germer observe with an amorphous nickel target?

Localizing matter waves and the wave-particle duality

The dilemma:

- ▶ Pure sine wave $\psi = e^{i(kz - \omega t)}$ repeats infinitely in space
- ▶ Real electrons have particle behavior, implying some definite position within some region Δz
- ▶ **How to explain localization with waves?** This requires creating a superposition of many waves with different k values

Wave packet

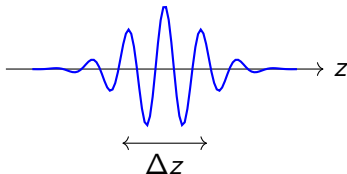
A **localized disturbance** formed by constructive interference of multiple waves with a spread of wavenumbers.

Single wave: k definite, position undefined
Wave packet: position localized, k has spread Δk

Building a wave packet

Superposition principle:

- ▶ Combine waves with wavenumbers spanning range Δk around central k_0
- ▶ Waves add **constructively** in localized region, **cancel** elsewhere
- ▶ Narrower spatial extent Δz requires broader spread Δk



Fundamental trade-off: Localization in space $\leftrightarrow$ spread in wavenumber

The space-wavenumber uncertainty

Fourier analysis of wave packets reveals a fundamental constraint:

- ▶ Product of spatial width and wavenumber spread is constant
- ▶ We cannot make both Δz and Δk arbitrarily small simultaneously

Spatial-Wavenumber Uncertainty

$$\Delta k \cdot \Delta z = 2\pi \quad (8)$$

Interpretation:

- ▶ Small Δz (well-localized) $\Rightarrow$ large Δk (many wavenumbers)
- ▶ Small Δk (well-known momentum) $\Rightarrow$ large Δz (spread out)

Uncertainty in momentum

Connecting wave and particle descriptions:

The de Broglie relation links wavenumber to momentum:

de Broglie's connection of momentum and wavenumber

$$p = \hbar k = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = \frac{h}{\lambda} \quad (9)$$

Key consequence:

- Uncertainty in k implies uncertainty in momentum:

$$\Delta p = \hbar \Delta k$$

- Substitute into $\Delta k \cdot \Delta z = 2\pi$ to obtain position-momentum relation

Heisenberg uncertainty principle

Starting point: $\Delta k \cdot \Delta z = 2\pi$ and $\Delta p = \hbar \Delta k$

Derivation:

$$\Delta k \cdot \Delta z = 2\pi \quad (10)$$

$$(\hbar \Delta k) \cdot \Delta z = 2\pi \hbar = h \quad (11)$$

$$\Delta p \cdot \Delta z = h \quad (12)$$

Heisenberg uncertainty principle

$$\boxed{\Delta p \cdot \Delta z = h} \quad (13)$$

Crucial insight: This is a **fundamental limit** on simultaneous knowledge of position and momentum—not a measurement limitation, but an intrinsic property of quantum reality.

Uncertainty is a fundamental property of matter

The uncertainty principle is NOT due to:

- ▶ Imperfect instruments or imprecise measurements
- ▶ Technological limitations

It IS about the nature of wave-particle duality:

- ▶ A particle *does not have* definite p and z simultaneously
- ▶ Wave nature makes simultaneous sharp values of p and z *impossible*

Physical consequence: Explains atomic stability

- ▶ If electron collapsed onto nucleus: $\Delta z \rightarrow 0$
- ▶ Then $\Delta p \rightarrow \infty$ — kinetic energy explodes
- ▶ Atoms find stable size balancing localization vs. kinetic cost

Phase and group velocity

A wave packet contains many frequency components

- ▶ Individual wave crests move at **phase velocity** $v_p = \omega/k$
- ▶ The envelope (packet shape) moves at **group velocity**
- ▶ These are generally *different* speeds!

Central question:

Which velocity represents the actual electron motion?

- ▶ Answer determines how we connect wave description to particle dynamics

Phase velocity is the wave motion

Definition: Phase velocity describes constant-phase surface propagation

For a plane wave $\psi = e^{i(kz - \omega t)}$, constant phase requires:

$$kz - \omega t = \text{const} \quad \Rightarrow \quad \boxed{v_p = \frac{\omega}{k}}$$

Key properties:

- ▶ Phase velocity can exceed speed of light c
- ▶ No violation of relativity: no information transfer at v_p
- ▶ In dispersive media, v_p depends on wavelength

Defining group velocity

Derivation from two-wave superposition:

Consider two waves with slightly different frequencies:

$$\psi_1 = \cos(kz - \omega t) \quad (14)$$

$$\psi_2 = \cos((k + dk)z - (\omega + d\omega)t) \quad (15)$$

Their sum creates a beat pattern (envelope):

$$\psi_1 + \psi_2 = 2 \cos\left(\frac{dk \cdot z - d\omega \cdot t}{2}\right) \cos(kz - \omega t)$$

Envelope moves where its argument is constant:

$$v_g = \frac{\partial \omega}{\partial k}$$

Group velocity is the derivative of the dispersion relation $\omega(k)$

Group velocity carries energy and information

Physical significance of v_g :

- ▶ Energy propagates in the wave packet *envelope*
- ▶ Information travels with the packet shape at v_g
- ▶ Individual crests passing through envelope carry no net energy

For electrons:

- ▶ Group velocity is equivalent to classical particle velocity
- ▶ $v_g = \partial\omega/\partial k$ connects wave and particle pictures

Common misconceptions in the uncertainty principle

Misconception 1: Uncertainty comes from measurement error

In reality: Uncertainty is **intrinsic** to quantum reality, not instrumental limitation

Misconception 2: phase velocity is the particle velocity

In reality: $v_p \neq$ particle velocity; **group velocity** v_g carries energy

Misconception 3: “sometimes wave, sometimes particle”

In reality: Electrons exhibit simultaneous wave and particle behavior

Summary: the foundation of wave-like behavior for electrons

Three pillars of electron-wave

$$\boxed{\lambda = \frac{h}{p}, \quad \Delta p \cdot \Delta z = h, \quad v_g = \frac{\partial \omega}{\partial k}} \quad (16)$$

- ▶ **de Broglie relation** $\lambda = h/p$: bridges particle momentum to wave interference
- ▶ **Uncertainty principle** $\Delta p \cdot \Delta z = h$: sets fundamental limits on localization
- ▶ **Group velocity** $v_g = \partial \omega / \partial k$: connects wave packets to electron dynamics

These relations form the foundation for our course of physics of materials